low_power_energy_characterization (Initial Draft)

Domain: Edge Iot | Document ID: low_power_energy_characterization_v1
Specification Standard: IEEE Capstone Evaluation Corpus

Executive Overview:

Measuring Joules-per-query on 15-watt single-board computers under continuous transformer inference loads.

Methodological Verification & System Lineage:

This specification is catalogued in IntentCloud for intent-aware hybrid retrieval.
Text chunks are indexed into 384-dimensional dense semantic vectors and 1,000,003-dimensional sparse feature-hashed token projections.
Candidate pools are unified using scale-invariant Reciprocal Rank Fusion (k=60).
Explainable citation attribution is extracted using cross-encoder cross-attention.